

# **SOFTWARE CHANGE IMPLEMENTATION REPORT**

## **Student Management System**

Change Request: CR-016

Show / Hide Password Visibility Toggle

JANSAN.J

Student ID: 24ICT08

Software Configuration Management - SCM (K72C001M12)

Version 1.0

08 October 2026

University College of Jaffna / Department of Interdisciplinary Studies

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# 1. Change Request

## 1.1 Change Request Identification

| Field                 | Details                                        |
|-----------------------|------------------------------------------------|
| Change Request ID     | CR-016                                         |
| Change Request Title  | Show / Hide Password Visibility Toggle         |
| Priority              | High                                           |
| Difficulty            | Easy                                           |
| Change Type           | Frontend UI / UX and Accessibility Enhancement |
| Implementation Status | Completed Successfully                         |
| Student / Developer   | C. Lavasujan (Student ID: 24ICT06)             |
| Date                  | 08 October 2026                                |

## 1.2 Business Requirement

The Student Management System requires a password visibility control so users can temporarily reveal entered credentials when needed. The primary purpose is to reduce typing errors and failed login attempts while keeping passwords masked by default.

As an extended requirement, the same show / hide behaviour is provided in the Admin Dashboard Add New Student modal so administrators can verify password entry while creating student accounts.

## 1.3 Existing Behaviour

- The password field is intended to protect credential entry by masking characters.
- Users may make typing errors when the password remains hidden, increasing the possibility of failed login attempts.
- The CR-016 implementation introduces an explicit visibility toggle for the login password field.
- The extended requirement also applies the same control to the Add New Student modal.

## **1.4 Requested Change**

- Add a client-side show / hide control to the Login page password field.
- Keep the default password state masked on component load.
- Allow the user to switch the input between masked (password) and visible (text) states.
- Update the toggle icon dynamically to represent the current visibility state.
- Ensure activating the toggle does not submit the login form.
- Provide keyboard-accessible and screen-reader-friendly toggle controls.
- Extend the same password visibility functionality to the Admin Dashboard Add Student modal.

## **1.5 Change Objective**

The objective of CR-016 is to improve credential-entry usability without weakening the default password-protection behaviour. The implementation provides controlled visibility on demand, retains masking by default, and preserves normal authentication and student-creation submission behaviour.

## **2. Change Analysis & Impact**

### **2.1 Change Scope**

#### In Scope

- Password visibility toggle on the Login page.
- Masked-by-default password input state.
- Visible text state after user activation.
- Dynamic Eye / EyeOff icon state.
- Toggle button configured as a non-submitting button.
- ARIA labeling and pressed-state support.
- Keyboard activation through a native button.
- Password visibility toggle in the Admin Dashboard Add Student modal.
- Regression verification of authentication and form submission behaviour.

#### Out of Scope

- Changes to backend authentication logic.
- Changes to password hashing.
- Changes to database schema or password storage.
- Changes to user authorization or role-based redirects.
- Changes to unrelated student-management functionality.

### **2.2 Assumptions and Constraints**

#### Assumptions

- The existing login authentication logic remains unchanged.
- The password value is already held by the form and only its input presentation is changed.
- The Add Student modal can reuse the same visibility-toggle interaction pattern.
- A native button remains available for keyboard and assistive-technology interaction.

#### Constraints

- The default state must remain masked.
- The toggle must not submit the form.
- Visible password text must only appear when the user deliberately activates the control.
- Password values must not be persisted in localStorage or sessionStorage.
- Existing validation and role-based redirect behaviour must continue to work.

## 2.3 Impact Analysis

| Area                           | Impact | Details                                                                                                                     |
|--------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------|
| Frontend                       | High   | LoginFormSection.jsx and StudentFormModal.jsx are enhanced with password visibility state and toggle controls.              |
| Backend                        | None   | No backend authentication or credential-processing changes are identified.                                                  |
| Database                       | None   | No database schema or password-storage changes are identified.                                                              |
| API                            | None   | No API contract or endpoint change is identified.                                                                           |
| Authentication / Authorization | Low    | Existing authentication flow and role-based redirects are preserved.                                                        |
| Security                       | Medium | The feature is controlled by explicit user action, retains masked-by-default behaviour, and avoids client-side persistence. |
| User Interface / Styling       | Medium | login.css and dashboard.css are updated for toggle positioning and hover/placement behaviour.                               |
| Configuration                  | None   | No environment or build configuration change is identified.                                                                 |
| External Services              | None   | No third-party or cloud service is affected.                                                                                |
| Performance                    | Low    | The change is a lightweight client-side state toggle with negligible processing overhead.                                   |
| Testing                        | Medium | Functional, accessibility, submission, authentication, and regression verification are applicable.                          |
| Documentation                  | Medium | This implementation report records the implementation, testing, evidence, and closure status.                               |



## 2.4 Affected Components and Files

| Component / File                             | Type               | Changed? | Purpose / Reason for Change                                               |
|----------------------------------------------|--------------------|----------|---------------------------------------------------------------------------|
| frontend/src/components/LoginFormSection.jsx | Frontend           | Yes      | Added showPassword state, input type switching, and visibility toggle UI. |
| frontend/src/styles/login.css                | Frontend / Styling | Yes      | Added positioning and hover styling for the password visibility control.  |
| frontend/src/components/StudentFormModal.jsx | Frontend           | Yes      | Extended password visibility control to the Add New Student modal.        |
| frontend/src/styles/dashboard.css            | Frontend / Styling | Yes      | Added styling support for the modal password toggle.                      |
| Backend authentication                       | Backend            | No       | Existing authentication logic is preserved.                               |
| Database / password storage                  | Database           | No       | No schema or storage change is identified.                                |

### 3. Requirements & Acceptance Criteria

#### 3.1 Functional Requirements

| ID    | Requirement                                                                | Status |
|-------|----------------------------------------------------------------------------|--------|
| FR-01 | Login password field supports masked and visible presentation.             | Met    |
| FR-02 | The password field is masked by default when the page loads.               | Met    |
| FR-03 | Clicking the visibility control switches the input to visible text.        | Met    |
| FR-04 | Activating the control again returns the field to masked mode.             | Met    |
| FR-05 | The visibility icon reflects the current masked / visible state.           | Met    |
| FR-06 | The visibility control does not submit the form prematurely.               | Met    |
| FR-07 | The control supports keyboard interaction.                                 | Met    |
| FR-08 | The control provides dynamic aria-label and aria-pressed information.      | Met    |
| FR-09 | Visible password text remains legible and is not obscured by the toggle.   | Met    |
| FR-10 | Valid credentials continue to authenticate successfully while visible.     | Met    |
| FR-11 | The Admin Dashboard Add Student modal provides the same visibility toggle. | Met    |

#### 3.2 Non-Functional Requirements

| ID     | Area            | Requirement                                                                               | Status |
|--------|-----------------|-------------------------------------------------------------------------------------------|--------|
| NFR-01 | Usability       | Password visibility can be changed on demand to reduce entry errors.                      | Met    |
| NFR-02 | Accessibility   | The toggle is keyboard accessible and exposes accessible state labels.                    | Met    |
| NFR-03 | Consistency     | The same interaction pattern is available in Login and Add Student contexts.              | Met    |
| NFR-04 | Maintainability | The feature uses localized frontend state without changes to authentication architecture. | Met    |

### 3.3 Security Requirements

| ID     | Security Requirement                                                         | Status |
|--------|------------------------------------------------------------------------------|--------|
| SEC-01 | Password remains masked by default.                                          | Met    |
| SEC-02 | Visible password text appears only after explicit user activation.           | Met    |
| SEC-03 | Password is not persisted in localStorage or sessionStorage.                 | Met    |
| SEC-04 | Existing authentication logic is preserved while input presentation changes. | Met    |
| SEC-05 | Form submission is not triggered by the visibility control.                  | Met    |

### 3.4 Acceptance Criteria

| ID     | Acceptance Criterion                                                  | Status |
|--------|-----------------------------------------------------------------------|--------|
| AC-001 | Users can toggle password visibility between hidden and visible.      | Met    |
| AC-002 | Default state of the password field on load is masked.                | Met    |
| AC-003 | Toggle icon updates dynamically to match visibility state.            | Met    |
| AC-004 | Activating the toggle does not trigger login form submission.         | Met    |
| AC-005 | Toggle button is keyboard accessible and includes aria-label.         | Met    |
| AC-006 | Password text remains legible and is not obscured by the toggle icon. | Met    |
| AC-007 | Valid credentials submitted in either state log in successfully.      | Met    |
| AC-008 | Existing validation errors and role-based redirects continue to work. | Met    |
| AC-009 | The Add Student modal supports password visibility toggling.          | Met    |

## 4. Risk & Configuration Management

### 4.1 Risk Analysis

| Risk                                 | Likelihood | Impact | Mitigation / Control                                                                            | Outcome   |
|--------------------------------------|------------|--------|-------------------------------------------------------------------------------------------------|-----------|
| Accidental form submission           | Medium     | Medium | Use type="button" on the visibility control so the toggle cannot act as a submit control.       | Mitigated |
| Password exposed by default          | Low        | High   | Initialize showPassword to false and retain the password input type by default.                 | Mitigated |
| Unreadable password text near icon   | Medium     | Medium | Add sufficient right-side padding to the input area.                                            | Mitigated |
| Accessibility failure                | Medium     | Medium | Use a native button with dynamic aria-label, aria-pressed, and keyboard activation support.     | Mitigated |
| Login regression                     | Low        | High   | Verify authentication with visible credentials and retain existing validation / redirect logic. | Mitigated |
| Inconsistent extended implementation | Medium     | Medium | Apply the same visibility behaviour to the Add Student modal and verify it separately.          | Mitigated |
| Credential persistence               | Low        | High   | Do not write password values to localStorage or sessionStorage.                                 | Mitigated |

### 4.2 Risk Mitigation Summary

The main risks are controlled at the frontend interaction level. The password remains masked by default, the toggle is explicitly non-submitting, accessibility metadata is provided, and the visible state is user-controlled. The implementation also extends the same control to the Add Student modal so the documented extended requirement is applied consistently.

### 4.3 Configuration Items

| CI ID | Configuration Item           | Type                  | Changed? | Status    |
|-------|------------------------------|-----------------------|----------|-----------|
| CI-01 | LoginFormSection.jsx         | Source Code           | Yes      | Modified  |
| CI-02 | login.css                    | Source Code / Styling | Yes      | Modified  |
| CI-03 | StudentFormModal.jsx         | Source Code           | Yes      | Modified  |
| CI-04 | dashboard.css                | Source Code / Styling | Yes      | Modified  |
| CI-05 | Authentication Logic         | Backend / Security    | No       | Unchanged |
| CI-06 | Database / Password Storage  | Database              | No       | Unchanged |
| CI-07 | CR-016 Implementation Report | Documentation         | Yes      | Modified  |

### 4.4 Baseline and Version Information

| Field                          | Details                                                                                                                   |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Repository / Project           | Student Management System                                                                                                 |
| Base Branch                    | JFICT06/CR006                                                                                                             |
| Starting Commit / Baseline ID  | CR006                                                                                                                     |
| Baseline Date                  | 08/10/2026                                                                                                                |
| Baseline Description           | Existing Student Management System login and Admin Dashboard functionality before CR-016 password visibility enhancement. |
| Working Branch                 | JFICT06/CR016                                                                                                             |
| Implementation Start Date      | 08/10/2026                                                                                                                |
| Implementation Completion Date | 09/10/2026                                                                                                                |

## 5. Implementation

### 5.1 Implementation Summary

CR-016 implements password visibility toggling in the Student Management System. The Login page now allows the user to move between masked and visible password presentation through a dedicated control. The same behaviour is also implemented in the Admin Dashboard Add Student modal as an extended requirement.

The source implementation uses React state to control the password input type, provides accessible state labels, prevents accidental form submission through the toggle control, and retains masked-by-default behaviour.

### 5.2 Technical Implementation

| Technical Area                 | Changed? | Implementation Details                                                                                                        |
|--------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------|
| Frontend                       | Yes      | LoginFormSection.jsx and StudentFormModal.jsx use showPassword state and switch input presentation between password and text. |
| Styling                        | Yes      | login.css and dashboard.css position and style the toggle control.                                                            |
| Backend                        | No       | No authentication controller or backend credential-processing modification is identified.                                     |
| Database                       | No       | No database schema or password-storage modification is identified.                                                            |
| API                            | No       | No request or response contract change is identified.                                                                         |
| Authentication / Authorization | No       | Existing authentication logic and role-based redirects remain unchanged.                                                      |
| Security                       | Yes      | Masked-by-default presentation, non-persistent visible state, and controlled user activation are preserved.                   |
| Accessibility                  | Yes      | Dynamic aria-label / aria-pressed attributes and native button keyboard support are implemented.                              |

### 5.3 Files Modified

| File                                         | Component          | Purpose of Modification                                                   |
|----------------------------------------------|--------------------|---------------------------------------------------------------------------|
| frontend/src/components/LoginFormSection.jsx | Login Form         | Added showPassword state, dynamic input type, and toggle button.          |
| frontend/src/styles/login.css                | Login Styling      | Added positioning and hover states for the password visibility control.   |
| frontend/src/components/StudentFormModal.jsx | Student Form Modal | Added the same show / hide password interaction to the Add Student modal. |
| frontend/src/styles/dashboard.css            | Dashboard Styling  | Added styling support for the modal toggle.                               |

### 5.4 Database Changes

Database Changes: No.

No database schema, password-storage, or student-model changes are identified in the supplied CR-016 implementation report. The change affects the presentation of an already-entered password value only.

### 5.5 API Changes

API Changes: No.

The supplied CR-016 implementation report identifies no new or modified API endpoints, request bodies, or response structures. Existing authentication and student creation flows continue to be used.

## 5.6 Authentication / Authorization / Security Changes

Authentication: Existing authentication logic is preserved.

Authorization: Existing role-based access and redirects are preserved.

Security Controls / Validation: The password remains masked by default, visible only after explicit user activation, and is not persisted in localStorage or sessionStorage. The visibility control is configured as type="button" so it does not cause premature submission.

## 5.7 Implementation Logic

Main Workflow: Password Visibility Toggle

1. On component mount, showPassword is false and the password input uses type="password".
2. When the user activates the toggle, the component updates showPassword to true and the input uses type="text".
3. The toggle icon changes to represent the current state; the implementation records Eye for hidden state and EyeOff for visible state.
4. Activating the control again resets showPassword to false and masks the password.
5. The same interaction pattern is available in the Admin Dashboard Add Student modal.

## 5.8 Error and Data Handling

- The visibility control does not change the underlying password value; it changes only the input presentation.
- The toggle uses type="button" to avoid triggering form submission.
- The source implementation does not persist plaintext credentials in localStorage or sessionStorage.
- Existing validation and authentication errors remain handled by the surrounding form logic.

## 5.9 Important Technical Decisions

- React state controls the input type so visibility can be changed without replacing the password value.
- Native button semantics are retained for keyboard accessibility.
- Dynamic ARIA state communicates whether the password is currently visible.
- Masking remains the default state to reduce accidental disclosure.
- The extended Add Student requirement follows the same interaction model for consistency.



## 6. Testing & Verification

### 6.1 Testing Strategy

- Functional testing
- Positive testing
- Negative / interaction testing
- Accessibility / keyboard testing
- Form submission testing
- Authentication testing
- UI regression testing
- Security / credential-persistence verification
- Extended requirement verification

### 6.2 Testing Environment

| Field                    | Details                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Environment              | Development / Test environment; exact environment details not specified in the supplied CR-016 implementation report. |
| Application Version      | Version 1.0                                                                                                           |
| Browser / Client         | Web browser; exact browser/version not specified in the supplied source.                                              |
| Database                 | No CR-016 database change identified.                                                                                 |
| API / Server Environment | No CR-016 backend/API change identified.                                                                              |

### 6.3 Test Data / Test Account

| Field         | Details                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| Test Context  | Login page password-entry workflow.                                                                                         |
| Admin Context | Admin Dashboard Add Student modal.                                                                                          |
| Credentials   | Test credentials were used for authentication verification; exact account details are not specified in the supplied report. |
| Permissions   | Administrator access is required for the Add Student modal; exact role/account identifier is not specified.                 |

### 6.4 Test Cases

| Test ID | Test Case                                            | Expected Result                                                    | Actual Result               | Status |
|---------|------------------------------------------------------|--------------------------------------------------------------------|-----------------------------|--------|
| TC-01   | Verify default state of password input on page load. | Password field characters are masked.                              | Masked on mount.            | Pass   |
| TC-02   | Verify clicking the toggle button.                   | Input switches to visible text; clicking again masks it.           | Toggles correctly.          | Pass   |
| TC-03   | Verify toggle icon dynamic updates.                  | Eye shown when hidden and EyeOff shown when visible.               | Icons update correctly.     | Pass   |
| TC-04   | Verify form submission prevention.                   | Clicking the toggle does not submit the form.                      | Form is not submitted.      | Pass   |
| TC-05   | Verify keyboard accessibility.                       | Tab focuses the control; Space / Enter activates it.               | Keyboard interaction works. | Pass   |
| TC-06   | Verify authentication with visible credentials.      | Valid credentials submit successfully while visible.               | Login succeeds.             | Pass   |
| TC-07   | Verify toggle in Admin Dashboard Add Student.        | Admin can toggle password visibility while creating a new student. | Toggles correctly.          | Pass   |

## 6.5 Test Summary

| Field               | Details |
|---------------------|---------|
| Total Test Cases    | 7       |
| Passed              | 7       |
| Failed              | 0       |
| Blocked             | 0       |
| Not Tested          | 0       |
| Pass Rate           | 100 %   |
| Overall Test Result | Passed  |

Summary: All seven test scenarios recorded in the supplied CR-016 implementation report passed, including default masking, visibility switching, icon updates, submission protection, keyboard accessibility, authentication, and the extended Add Student implementation.

## 6.6 Regression Verification

Regression verification confirms that authentication logic remains functional when the password is visible, that the visibility control does not trigger form submission, and that the Admin Dashboard Add Student modal supports the extended requirement without changing the underlying student-creation flow.

The source report also states that existing validation errors and role-based redirects continue to work correctly.

## 6.7 Acceptance Criteria Verification

| Acceptance Criterion | Verification | Result | Evidence Reference          |
|----------------------|--------------|--------|-----------------------------|
| AC-001               | TC-02        | Met    | Screenshot 1 / Screenshot 2 |
| AC-002               | TC-01        | Met    | Screenshot 1                |
| AC-003               | TC-03        | Met    | Screenshot 2                |
| AC-004               | TC-04        | Met    | TC-04                       |
| AC-005               | TC-05        | Met    | TC-05                       |

|        |                                |     |                          |
|--------|--------------------------------|-----|--------------------------|
| AC-006 | UI inspection / implementation | Met | Screenshot 2             |
| AC-007 | TC-06                          | Met | TC-06                    |
| AC-008 | Regression verification        | Met | TC-06 / regression check |
| AC-009 | TC-07                          | Met | Screenshot 3             |

## 7. Traceability & Evidence

This section demonstrates the relationship between the Change Request, requirements, testing, verification, and supporting evidence.

### 7.1 Change Traceability

| Requirement / AC | Test Case(s)  | Evidence ID                                | Status |
|------------------|---------------|--------------------------------------------|--------|
| FR-01 - FR-05    | TC-01 - TC-03 | Evidence A-01 / B-01 / B-02                | Met    |
| FR-06 - FR-08    | TC-04 - TC-05 | Test Evidence B-04 / implementation record | Met    |
| FR-09 - FR-10    | TC-06         | Test Evidence B-05                         | Met    |
| FR-11            | TC-07         | Test Evidence B-06 / Screenshot 3          | Met    |
| SEC-01 - SEC-05  | TC-01 - TC-06 | Evidence A-01 / B-01 - B-05                | Met    |
| AC-001 - AC-009  | TC-01 - TC-07 | Screenshots 1 - 3 and test cases           | Met    |

Traceability Chain: Change Request -> Requirement -> Test -> Evidence -> Verification

## 7.2 Embedded Evidence

The user-specified screenshot evidence consists of three interface states. The actual image files were not included with the current source package, so this editable report reserves insertion areas for the captured screenshots rather than fabricating images.

Evidence A-01

Description: Login page with masked password showing the default state.

Related Requirement / Test: FR-02, AC-002 / TC-01

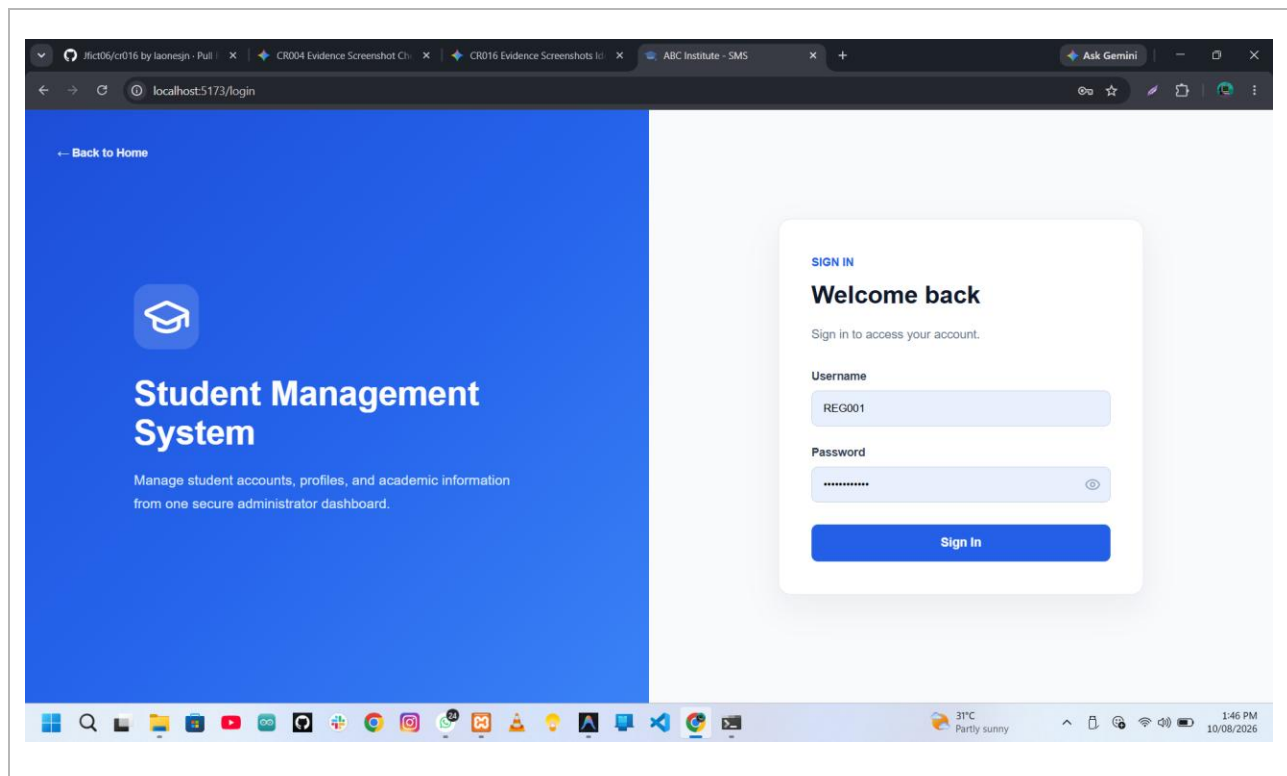


Figure 1: Login Page - Masked Password

## Evidence A-02

Description: Login page with visible password after activating the toggle.

Related Requirement / Test: FR-01, FR-03, FR-04, FR-05, AC-001, AC-003 / TC-02, TC-03

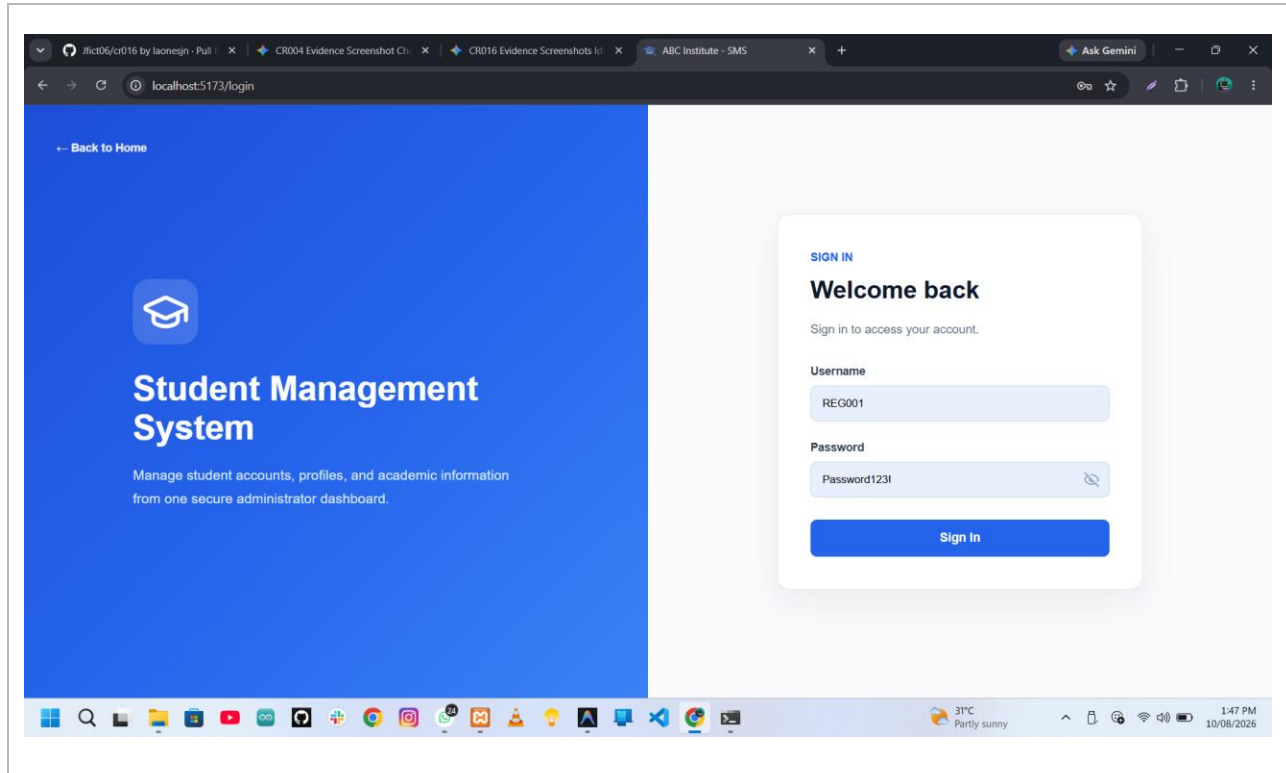


Figure 2: Login Page - Visible Password

## Evidence A-03

Description: Admin Dashboard Add Student modal with a visible password, demonstrating the extended requirement.

Related Requirement / Test: FR-11, AC-009 / TC-07

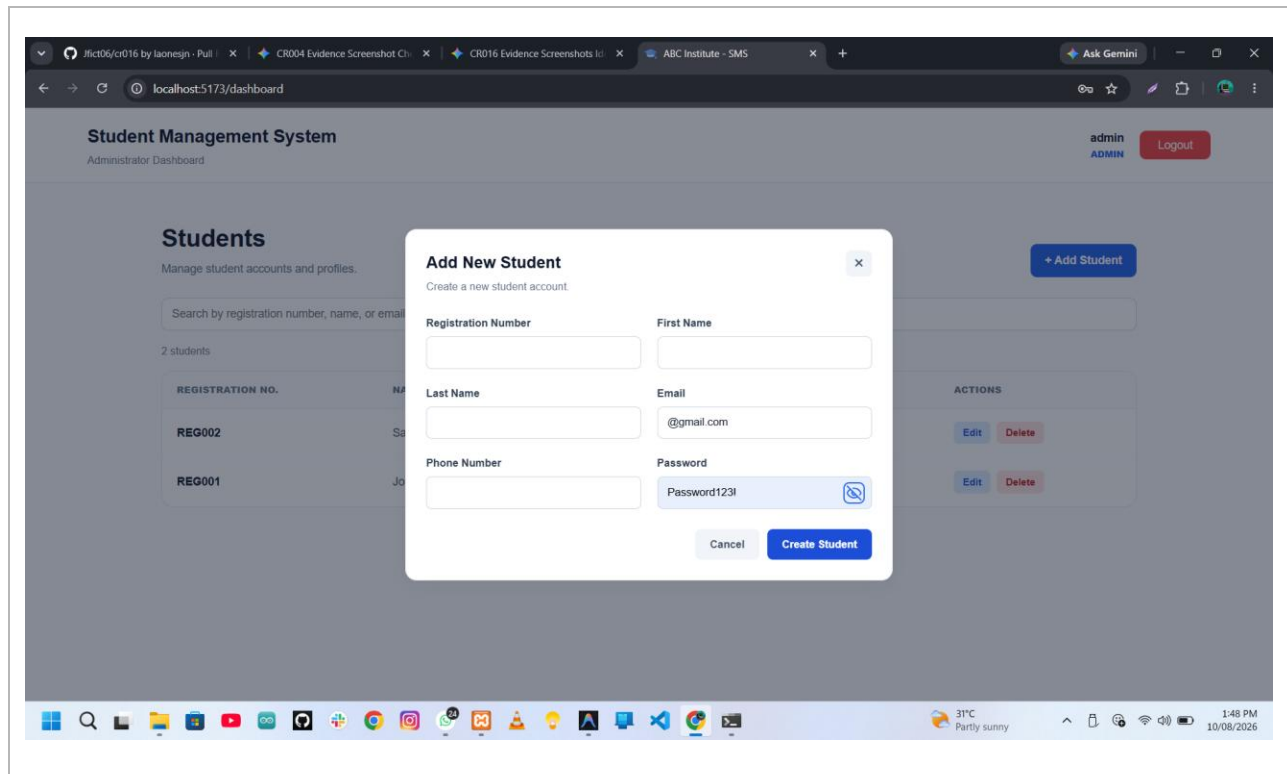


Figure 3: Admin Dashboard Add Student Modal - Visible Password

## 8. Version Control & Change Control

### 8.1 Repository and Branch Information

| Field             | Details                                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Repository        | <a href="https://github.com/sangeerththanan-academic/student-management-system.git">https://github.com/sangeerththanan-academic/student-management-system.git</a> |
| Source Branch     | JFICT016/CR004                                                                                                                                                    |
| Target Branch     | JFICT08/CR016                                                                                                                                                     |
| Pull Request Date | 08/10/2026                                                                                                                                                        |
| Review Status     | Pending                                                                                                                                                           |
| Merge Status      | Done                                                                                                                                                              |



## 8.2 Code Review / Change Review

| Review Comment                 | Action Taken                                                             | Commit ID     | Status              |
|--------------------------------|--------------------------------------------------------------------------|---------------|---------------------|
| Verify default masked state    | Initialize visibility state to false.                                    | Not specified | Resolved / Verified |
| Verify non-submitting toggle   | Use type="button" on visibility control.                                 | Not specified | Resolved / Verified |
| Verify accessibility           | Use dynamic aria-label and aria-pressed attributes with a native button. | Not specified | Resolved / Verified |
| Verify readability near toggle | Use padding-right so password text remains legible.                      | Not specified | Resolved / Verified |
| Verify extended requirement    | Implement the same toggle in the Add Student modal.                      | Not specified | Resolved / Verified |

## 8.3 Change Control Outcome

The supplied CR-016 implementation report records the change as completed successfully with no outstanding implementation issues. Repository, branch, commit, Pull Request, and formal reviewer details were not included in the source material and are therefore explicitly marked as not specified rather than being inferred.

## 9. Final Status & Closure

### 9.1 Final Change Status

| Item                   | Status                                                         |
|------------------------|----------------------------------------------------------------|
| Change Request ID      | CR-016                                                         |
| Change Request Title   | Show / Hide Password Visibility Toggle                         |
| Implementation Status  | Completed Successfully                                         |
| Testing Status         | Passed                                                         |
| Acceptance Criteria    | All Met                                                        |
| Demonstration / Review | Evidence spaces prepared; formal review metadata not specified |
| Final Acceptance       | Ready for submission / formal review                           |
| Final Change Status    | Implementation Complete / Evidence Ready                       |

### 9.2 Change Scope Summary

| Area                           | Changed?                          |
|--------------------------------|-----------------------------------|
| Frontend                       | Yes                               |
| Backend                        | No                                |
| Database                       | No                                |
| API                            | No                                |
| Authentication / Authorization | No - existing behaviour preserved |
| Security                       | Yes                               |
| Configuration                  | No                                |
| External Services              | No                                |
| Documentation                  | Yes                               |

### 9.3 Final Summary

CR-016 implements a controlled show / hide password visibility feature in the Student Management System. The Login page supports masked-by-default credentials, user-triggered visibility, dynamic icon feedback, accessible controls, and keyboard activation. Valid credentials continue to authenticate successfully when visible. The same visibility functionality is also implemented in the Admin Dashboard Add Student modal as the documented extended requirement.

The supplied implementation report records seven test cases as passed and all eight original acceptance criteria as satisfied; this report additionally records the extended Add Student criterion as met. No backend, database, or API changes are identified.

### 9.4 Outstanding Issues / Limitations

- No functional implementation defect is identified in the supplied source report.
- The actual three screenshot image files are not included in the current document package and should be inserted into the prepared evidence areas.
- Repository, branch, commit, Pull Request, and reviewer metadata were not specified in the supplied implementation report.

### 9.5 Follow-up Actions

| Action                                                                                          | Responsible Person | Status |
|-------------------------------------------------------------------------------------------------|--------------------|--------|
| Insert Screenshot 1 - masked Login password                                                     | C. Lavasujan       | Done   |
| Insert Screenshot 2 - visible Login password                                                    | C. Lavasujan       | Done   |
| Insert Screenshot 3 - visible Add Student password                                              | C. Lavasujan       | Done   |
| Complete repository / branch / commit / review metadata when authoritative values are available | C. Lavasujan       | Done   |
| Perform final document review                                                                   | C. Lavasujan       | Done   |

## 9.6 Closure Checklist

| Closure Item                           | Status    |
|----------------------------------------|-----------|
| Change requirements reviewed           | Completed |
| Change analysis completed              | Completed |
| Impact analysis completed              | Completed |
| Assumptions and constraints documented | Completed |
| Risk analysis completed                | Completed |
| Configuration items identified         | Completed |
| Implementation completed               | Completed |
| Test cases completed                   | Completed |
| Acceptance criteria verified           | Completed |
| Traceability completed                 | Completed |
| Evidence spaces prepared               | Completed |
| Version-control information recorded   | Done      |
| Review / Pull Request completed        | Done      |
| Regression verification completed      | Completed |
| Documentation completed                | Completed |
| Outstanding issues recorded            | Completed |
| Follow-up actions recorded             | Completed |
| Change ready for closure               | Completed |

## 9.7 Closure Comments

CR-016 has been documented in the same organizational and presentation style used for the supplied CR-007 implementation report. The report preserves the source terminology, records all seven supplied test cases as passed, and provides dedicated evidence areas for the three requested screenshots.

## 9.8 Final Closure

| Field               | Details                                                                                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Final Change Status | Implementation Complete - Evidence Ready                                                                                                                                |
| Closed By           | JANSAN.J (Student ID: 24ICT08)                                                                                                                                          |
| Closure Date        | 08 October 2026                                                                                                                                                         |
| Confirmation        | This report records the CR-016 implementation, testing, impact analysis, evidence structure, and current closure readiness based on the supplied implementation report. |
| Student Name        | JANSAN.J                                                                                                                                                                |
| Signature           | _____                                                                                                                                                                   |
| Date                | 08 October 2026                                                                                                                                                         |

## Appendix A - Additional Evidence

Supporting technical material confirming the implementation and verification results.

Evidence A-01

Description: Client-side password visibility implementation coverage.

Related Requirement / Test: FR-01 - FR-11 / TC-01 - TC-07

| Implementation Check                               | Recorded Result |
|----------------------------------------------------|-----------------|
| showPassword state controls masked / visible state | Pass            |
| Input type switches between password and text      | Pass            |
| Toggle button configured as type="button"          | Pass            |
| aria-label and aria-pressed state present          | Pass            |
| Keyboard activation via native button              | Pass            |
| No localStorage / sessionStorage persistence       | Pass            |
| Add Student modal extension                        | Pass            |

Evidence A-02

Description: Security and usability verification summary.

Related Requirement / Test: SEC-01 - SEC-05 / TC-01 - TC-06

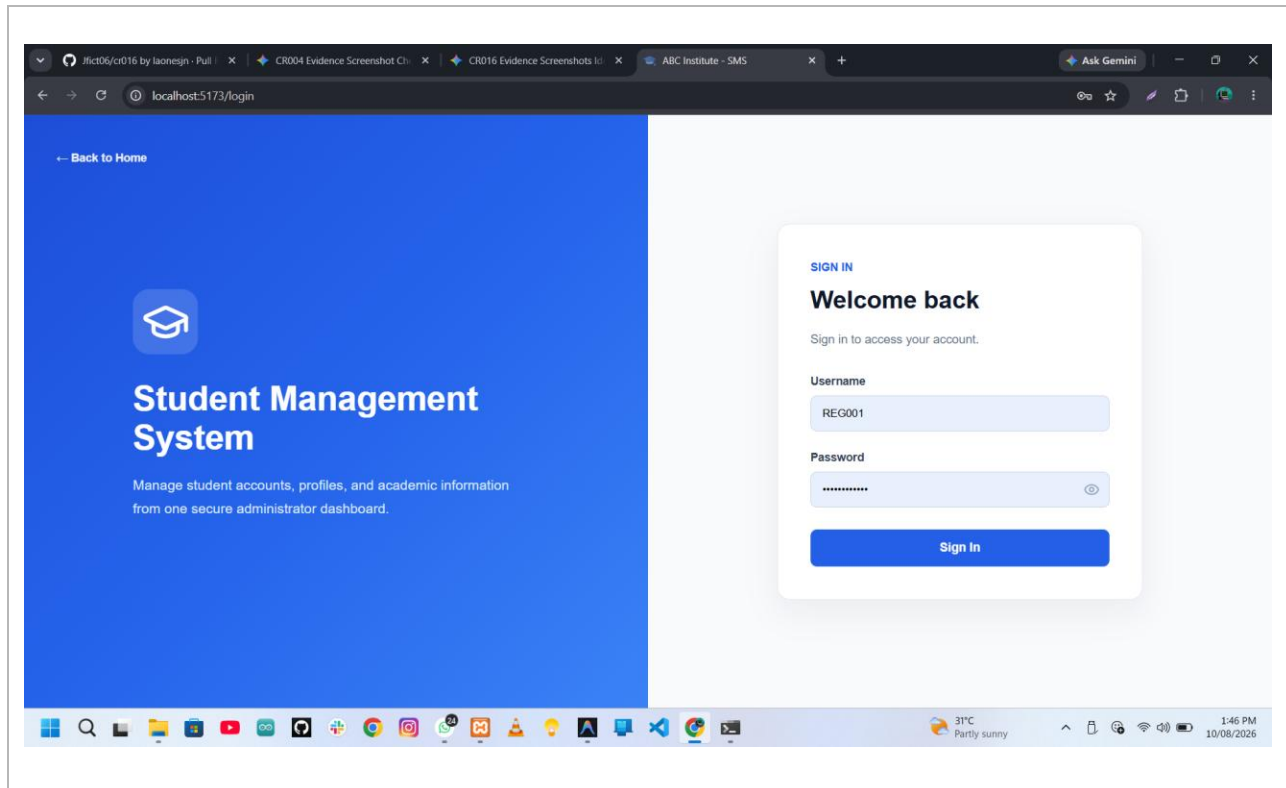
| Control                        | Result   |
|--------------------------------|----------|
| Masked by default              | Verified |
| User-controlled visibility     | Verified |
| No premature submission        | Verified |
| Authentication while visible   | Verified |
| No browser storage persistence | Verified |

## Appendix B - Screenshot Evidence

Replace each placeholder below with the corresponding screenshot captured for CR-016. Keep the mapped-criteria line so the evidence remains traceable.

### Figure 1: Screenshot 1 - Login Page with Masked Password

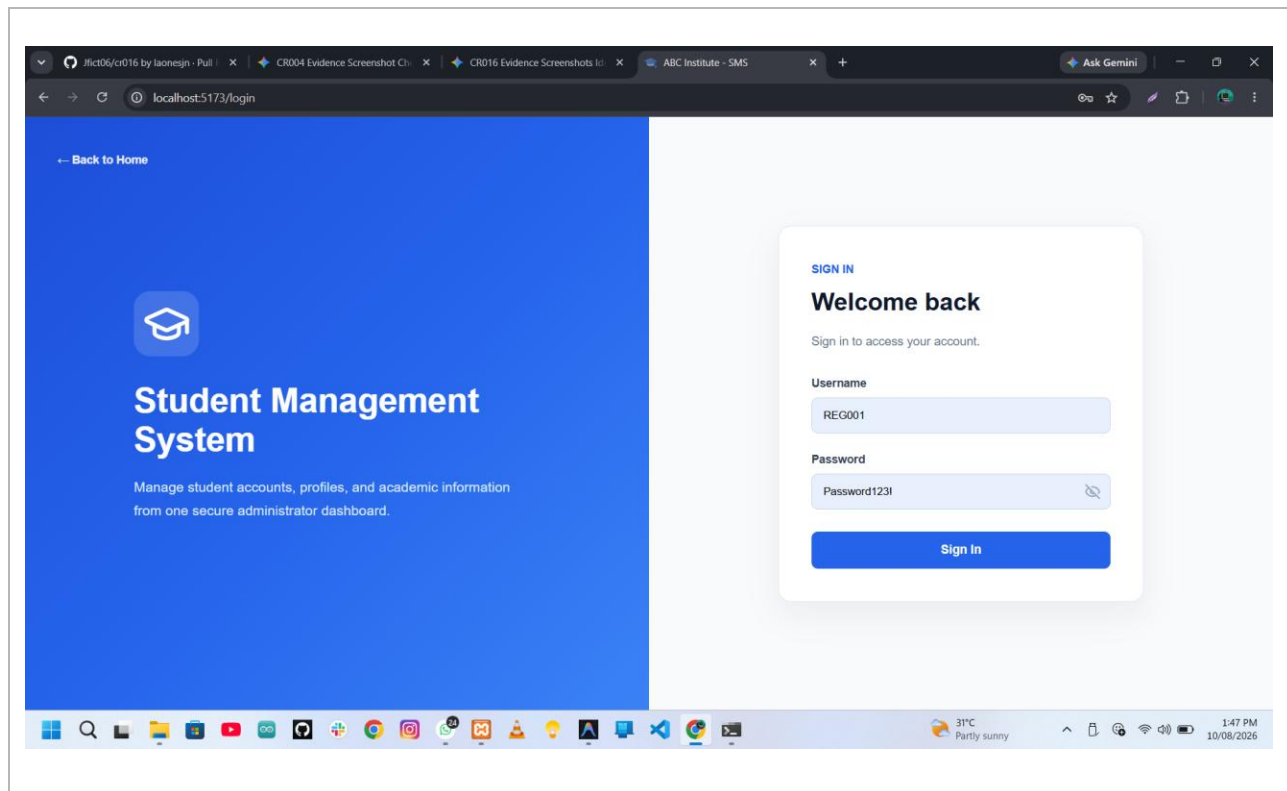
Mapped Criteria: AC-002 / TC-01



Observation: Default password state with characters hidden.

## Figure 2: Screenshot 2 - Login Page with Visible Password

Mapped Criteria: AC-001, AC-003, AC-007 / TC-02, TC-03, TC-06

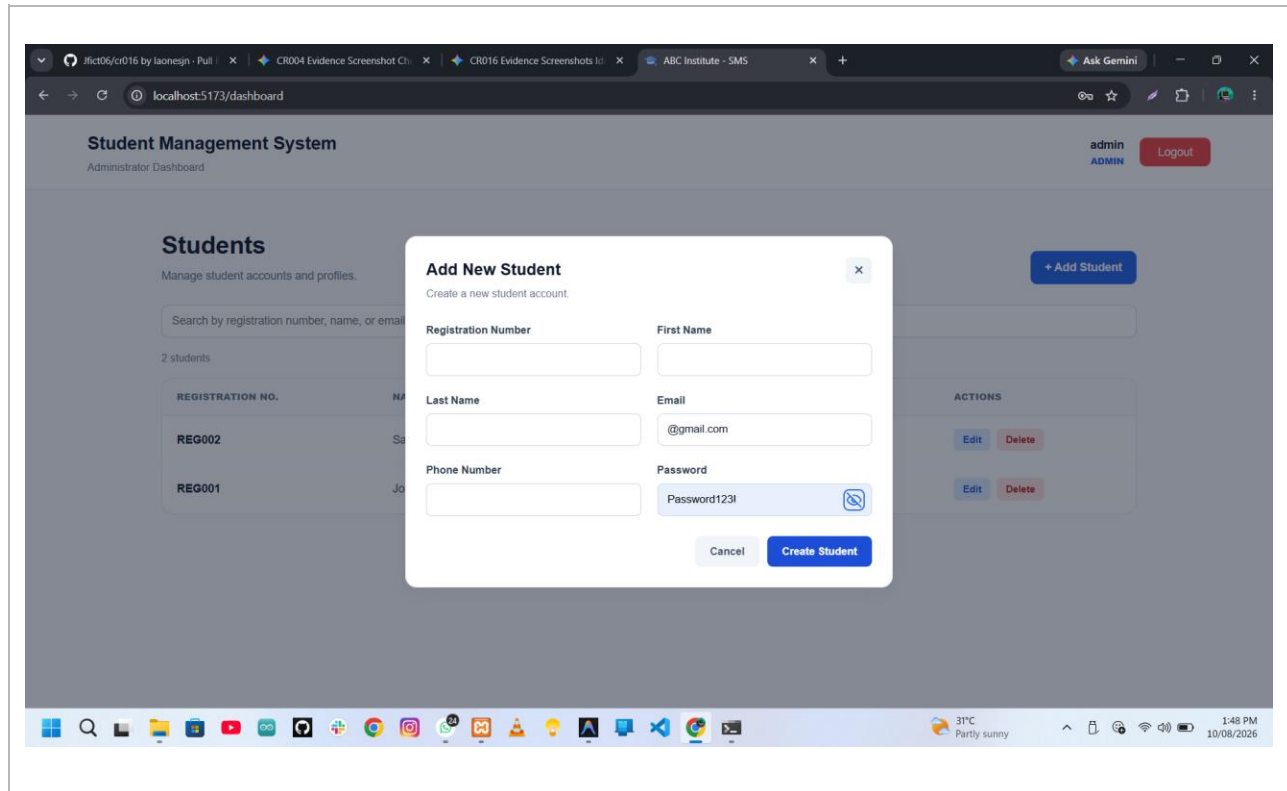


Observation: Password text visible after the user activates the toggle.



**Figure 3: Screenshot 3 - Admin Dashboard Add Student Modal with Visible Password**

Mapped Criteria: AC-009 / TC-07



Observation: Visible password state in the Add New Student modal demonstrating the extended requirement.

## Appendix C - Original Change Request Reference

The supplied CR-016 implementation report is the source basis for the implementation details, test results, security notes, and screenshot requirements recorded in this report.

| Change Request Element      | CR-016 Reference                                                                    |
|-----------------------------|-------------------------------------------------------------------------------------|
| Change Request ID           | CR-016                                                                              |
| Primary Feature             | Show / Hide Password Visibility Toggle                                              |
| Main Frontend Component     | frontend/src/components/LoginFormSection.jsx                                        |
| Main Styling File           | frontend/src/styles/login.css                                                       |
| Extended Frontend Component | frontend/src/components/StudentFormModal.jsx                                        |
| Extended Styling File       | frontend/src/styles/dashboard.css                                                   |
| Default State               | Masked                                                                              |
| Visible State               | Text / unmasked after toggle activation                                             |
| Accessibility               | Dynamic aria-label and aria-pressed; native button keyboard support                 |
| Security                    | No localStorage / sessionStorage persistence; masked by default                     |
| Testing                     | 7 supplied test cases, all Pass                                                     |
| Acceptance Criteria         | 8 supplied criteria, all Met; extended Add Student criterion also documented as Met |

### Evidence Requirements Referenced

- Screenshot 1: Login page with masked password.
- Screenshot 2: Login page with visible password.
- Screenshot 3: Admin dashboard Add Student modal with visible password.

## STUDENT REPORT COMPLETION CHECK

| Completion Item                                                              | Status    |
|------------------------------------------------------------------------------|-----------|
| Original Change Request has been reviewed.                                   | Completed |
| Business requirement has been documented.                                    | Completed |
| Current behaviour has been documented.                                       | Completed |
| Requested change has been documented.                                        | Completed |
| Scope has been identified.                                                   | Completed |
| Assumptions and constraints have been documented.                            | Completed |
| Impact analysis has been completed.                                          | Completed |
| Affected components have been identified.                                    | Completed |
| Affected files have been identified.                                         | Completed |
| Configuration items have been recorded.                                      | Completed |
| Baseline / version-control limitations have been recorded where unspecified. | Completed |
| Risks and mitigations have been documented.                                  | Completed |
| Requirements have been mapped to implementation.                             | Completed |
| Implementation details have been documented.                                 | Completed |
| Technical logic has been explained.                                          | Completed |
| Error and data handling have been documented.                                | Completed |
| Testing strategy has been defined.                                           | Completed |
| Test cases have been completed.                                              | Completed |
| Test results have been recorded.                                             | Completed |
| Acceptance criteria have been verified.                                      | Completed |
| Traceability has been completed.                                             | Completed |
| Evidence areas have been prepared.                                           | Completed |

|                                                               |           |
|---------------------------------------------------------------|-----------|
| Version-control / Pull Request information has been prepared. | Done      |
| Regression verification has been addressed.                   | Completed |
| Final change status has been determined.                      | Completed |
| Change ready for closure.                                     | Done      |